

Fuzzifying Derivatives of Regular Expressions

Besik Dundua^{1,2}, Furio Honsell³, Mariam Katamashvili^{1,2}, David Maisuradze¹

¹*Kutaisi International University, Kutaisi, Georgia*

²*VIAM, Tbilisi State University, Tbilisi, Georgia*

³*University of Udine, Udine, Italy*

There is growing interest in computer science and artificial intelligence in moving from *exact*, *qualitative* settings to *quantitative*, and ultimately *fuzzy*, ones. In this spirit, we extend the theory of finite automata, based on exact symbol equality, to fuzzy symbol matching. Following similarity-based reasoning [7], we enrich the alphabet with a *similarity relation* $\mathcal{R}$ and perform matching up to a fixed *cut* value $\mu \in (0, 1]$: two symbols match when their similarity is at least μ . Here $\mathcal{R}$ takes values in $[0, 1]$ endowed with a left-continuous T-norm (a residuated lattice [4]), and is reflexive, symmetric, and transitive.

We generalize the theory of Brzowski derivatives [1], revisited in [5], which constructs automata directly from regular expressions: after reading a word, the current state is the corresponding derivative of the expression. The similarity relation on symbols is extended to regular expressions, words, and languages, yielding in each case a similarity relation. Since intersection and complement do not fuzzify well, our results concern the *restricted regular expressions*: empty language, empty word, letters, union, concatenation, and Kleene star.

We introduce *fuzzy language and expression derivatives*, in which each symbol of the input word is matched up to the cut value μ , and prove that they preserve regularity. We further establish a *semantic-syntactic correspondence* (the language of the fuzzy derivative of an expression coincides with the fuzzy derivative of its language) and a *finiteness* result: iterating fuzzy derivatives over all words yields only finitely many distinct languages.

One of the main results connects our fuzzy theory to the classical, crisp one. For a fixed cut value μ and Gödel's minimum T-norm, the similarity relation induces an equivalence on the alphabet, whose quotient we call the *quotient alphabet* Σ^μ . We prove that ordinary Brzowski derivative automata over Σ^μ capture exactly the fuzzy derivatives over the original alphabet. Most of these results are formalized in the Rocq prover [6], see [3], building on the Coquand-Siles [2] formalization of regular-expression equivalence via Brzowski derivatives. Furthermore, we explore how to relate our similarity-based approach to the fuzzy-set construction of Stamenković and Ćirić [8].

References

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